CLASS 9 · CBSE · GANITA MANJARI PART 2 · FORMULA BANK

Chapter 13: Two Variables, One Line

Every formula and result of the chapter on one sheet

1	Standard form	$ax + by + c = 0$, with a and b not both zero
2	Number of solutions of one equation	Infinitely many; they form a straight line
3	Slope through two points	$\frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{Rise}}{\text{Run}}$
4	Slope-intercept form	$y = mx + d$, with slope m and y-intercept d
5	From standard form	$m = -\frac{a}{b}$ and $d = -\frac{c}{b}$
6	Horizontal line	$y = k$, slope 0
7	Vertical line	$x = k$, slope undefined
8	Line through the origin	$ax + by = 0$
9	Unique solution	$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ (intersecting lines)
10	Infinitely many solutions	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ (coincident lines)
11	No solution	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ (parallel lines)
12	Two-digit number	$10x + y$, with tens digit x and units digit y
13	Substitution method	Express one variable from one equation and put it in the other
14	Elimination method	Make one coefficient equal in both equations, then add or subtract